

Lecture 8: Control Unit (CU)

Learning Objectives

By the end of this lecture, students will be able to:

- Define the Control Unit (CU).
 - Explain the role of the Control Unit inside the CPU.
 - Understand how the CU coordinates CPU operations.
 - Differentiate between Hardwired and Microprogrammed Control Units.
 - Explain the purpose of Program Counter (PC), Instruction Register (IR), Memory Address Register (MAR), and Memory Data Register (MDR/MBR).
 - Understand the role of control signals.
 - Explain how the Control Unit works during instruction execution.
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1. Introduction

Teaching Script

First 30-Second SEO Phrase

Naturally include:

"In today's Computer Organization and Assembly Language lecture, we'll study the Control Unit, Program Counter (PC), Instruction Register (IR), Memory Address Register (MAR), Memory Data Register (MDR), control signals, and the difference between Hardwired and Microprogrammed Control Units."

In the previous lecture, we learned about the Arithmetic Logic Unit (ALU) and saw how it performs arithmetic and logical operations.

But here's an important question:

Who tells the ALU what operation to perform?

Who decides when to fetch an instruction from memory?

Who decides when data should move into a register?

Who tells the CPU when to read from memory or write to memory?

The answer is the **Control Unit**, or CU.

The Control Unit is like the manager of the CPU. It doesn't perform calculations itself, but it coordinates every activity inside the processor.

In today's lecture, we'll study the Control Unit, its internal working, different types, important CPU registers related to instruction execution, and how it controls the entire execution process.

Let's begin.

2. What is the Control Unit?

Definition

The Control Unit (CU) is the component of the CPU that directs and coordinates all operations by generating control signals and managing the execution of instructions.

In simple words:

The Control Unit tells every part of the computer what to do and when to do it.

Unlike the ALU, the Control Unit does **not** perform arithmetic calculations.

Instead, it:

- Fetches instructions
- Decodes instructions
- Generates control signals
- Controls data movement

- Coordinates communication between CPU, memory, and input/output devices
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3. Why is the Control Unit Needed?

Imagine a classroom.

- Students represent the **ALU**, **Registers**, and **Memory**.
- The teacher represents the **Control Unit**.

Without the teacher:

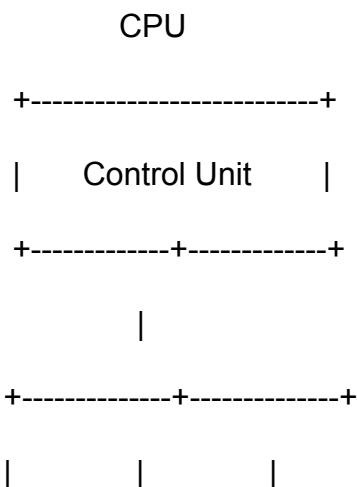
- Everyone works randomly.
- No one knows what to do.

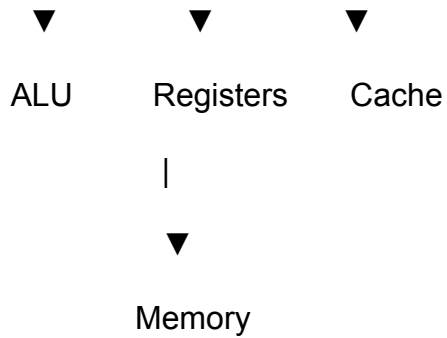
With the teacher:

- Instructions are clear.
- Everyone performs the correct task at the correct time.

Similarly, the Control Unit ensures that all CPU components work in the correct sequence.

4. Position of the Control Unit





The Control Unit communicates with almost every component inside the CPU.

5. Responsibilities of the Control Unit

The Control Unit performs several important tasks:

1. Fetches Instructions

Retrieves the next instruction from memory.

2. Decodes Instructions

Determines what operation the instruction specifies.

3. Generates Control Signals

Produces signals that coordinate operations across the CPU.

Examples:

- Read
- Write
- Load
- Increment
- Enable

4. Coordinates Components

Ensures that the ALU, registers, memory, and buses work together correctly.

5. Controls Data Movement

Moves data between:

- Memory ↔ Registers
 - Registers ↔ ALU
 - ALU ↔ Registers
-

6. Control Signals

The Control Unit communicates using **control signals**.

Think of them as commands.

Example:

Read Memory

↓

Load Register

↓

Perform Addition

↓

Store Result

↓

Next Instruction

Each signal triggers a specific hardware action.

7. Important CPU Registers Used by the Control Unit

The Control Unit relies on several special-purpose registers.

We'll study four of the most important ones.

A. Program Counter (PC)

Definition

The Program Counter stores the address of the next instruction to be executed.

Example:

Memory

1000 → LOAD A

1001 → ADD B

1002 → STORE C

1003 → HALT

Initially:

PC = 1000

After fetching the instruction:

PC = 1001

The PC automatically updates to point to the next instruction unless a branch or jump changes it.

B. Instruction Register (IR)

Definition

The Instruction Register stores the instruction currently being executed.

Example:

LOAD A

This instruction is fetched from memory and placed into the IR.

The Control Unit reads the IR to determine what operation to perform.

C. Memory Address Register (MAR)

Definition

The Memory Address Register stores the address of the memory location that the CPU wants to access.

Example:

Read memory location 2050

The MAR contains:

2050

The memory hardware uses this address to locate the requested data or instruction.

D. Memory Data Register (MDR)

Also called the **Memory Buffer Register (MBR)** in some textbooks.

Definition

The Memory Data Register temporarily stores data being transferred to or from memory.

Example:

Memory location 2050 contains:

50

The value:

50

is loaded into the MDR before being used by the CPU.

8. Interaction of These Registers

Program Counter

|

▼

MAR

|

▼

Memory

|

▼

MDR

|

▼

IR

|



Control Unit

This sequence is repeated for every instruction.

9. Types of Control Units

There are two main types.

A. Hardwired Control Unit

A Hardwired Control Unit uses fixed digital circuits to generate control signals.

Characteristics

- Very fast
- Difficult to modify
- Hardware-based
- Common in RISC processors

Advantages

- High speed
- Efficient execution

Disadvantages

- Not flexible
 - Difficult to redesign if the instruction set changes
-

B. Microprogrammed Control Unit

A Microprogrammed Control Unit stores control information in a special memory called **Control Memory**.

Instead of fixed hardware logic, it executes sequences of **microinstructions** to generate control signals.

Characteristics

- Easier to modify
- More flexible
- Common in many complex instruction set (CISC) designs

Advantages

- Easy to update
- Simpler to implement complex instruction sets

Disadvantages

- Generally slower than hardwired control because it must fetch microinstructions
-

10. Hardwired vs Microprogrammed

Feature	Hardwired CU	Microprogrammed CU
Speed	Faster	Slower
Flexibility	Low	High
Design	Fixed hardware	Control memory + microinstructions

Modification Difficult

Easier

Common
Use Many RISC
 processors

Many CISC processors

11. How the Control Unit Executes an Instruction

Suppose the instruction is:

ADD A, B

Execution steps:

Step 1

The PC holds the address of the next instruction.

↓

Step 2

The address is copied to the MAR.

↓

Step 3

Memory sends the instruction to the MDR.

↓

Step 4

The instruction is copied to the IR.



Step 5

The Control Unit decodes the instruction.



Step 6

The Control Unit sends signals to the ALU.



Step 7

The ALU performs the addition.



Step 8

The result is stored.



Step 9

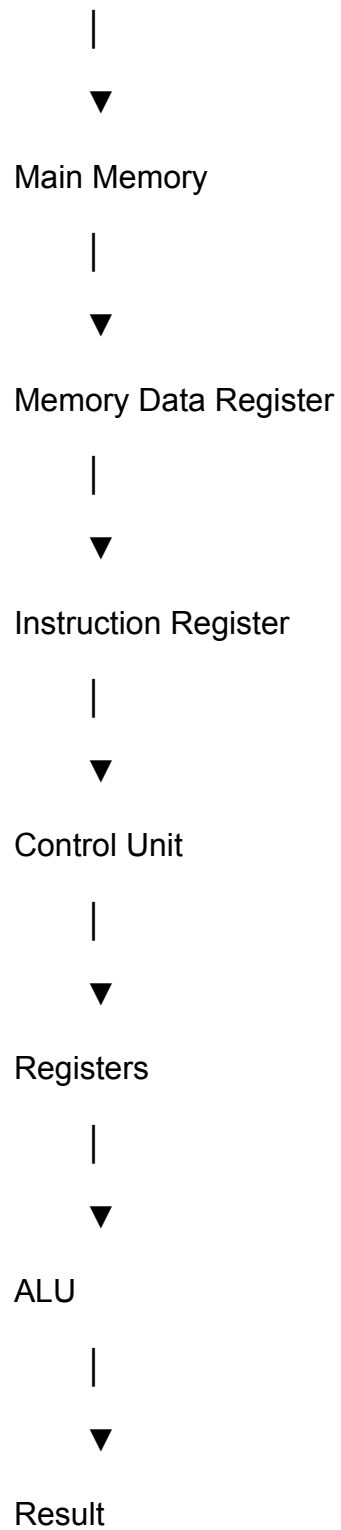
The PC advances to the next instruction.

12. Complete Flow Diagram

Program Counter



Memory Address Register



13. Real-Life Analogy

Imagine a restaurant.

Customer places an order.

↓

Manager receives the order.

↓

Manager tells the chef what to cook.

↓

Chef prepares the food.

↓

Waiter delivers the food.

In this analogy:

- Customer → Program
- Manager → Control Unit
- Chef → ALU
- Ingredients → Memory
- Tray → Registers

The manager doesn't cook but ensures everything happens in the correct order.

14. Practical Example

Consider this C++ statement:

```
int c = a + b;
```

A simplified execution sequence:

1. The instruction is fetched from memory.
 2. The Control Unit decodes it.
 3. Values of **a** and **b** are loaded into registers.
 4. The Control Unit instructs the ALU to add them.
 5. The result is stored in a register.
 6. The result is written back to memory.
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15. Relationship Between the Control Unit and the ALU

Control Unit



Reads instruction



Generates control signals



ALU performs operation



Registers store result



Memory stores final value

The Control Unit directs; the ALU computes.

Lecture Summary

Today, we studied the Control Unit (CU), the component responsible for coordinating every operation inside the CPU.

We learned how the Control Unit fetches instructions, decodes them, generates control signals, and manages communication between the ALU, registers, memory, and input/output devices.

We also explored the functions of the Program Counter (PC), Instruction Register (IR), Memory Address Register (MAR), and Memory Data Register (MDR), which play essential roles during instruction execution.

Finally, we compared Hardwired and Microprogrammed Control Units and understood the advantages and disadvantages of each.

In the next lecture, we'll study **CPU Registers** in greater detail, including general-purpose and special-purpose registers, their functions, and how they improve processor performance.

Thank you for watching, and I'll see you in the next lecture.
